

Assignment 3

CSE 350

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sec: 1 (A)

V_{in}	V_{out}
0.01	2.5
0.1	5
0.18	2.5
1.00	10
1.01	12.5
1.10	15
1.11	17.5

Ans to the ques no-1

$$I =$$

Q) $V_{OH} = 10V$

$V_{OL} = 0.2V$

Case - I

low level output (0) $\rightarrow$ high level input (1) (d)

$V_{OL} = 0.2V$ low level output

Supply current from driver $= \frac{12 - 0.2}{2.2} = 5.3636 \text{ mA}$

Demand current $= \frac{0.2 - (-12)}{15 + 100} = 0.106 \text{ mA}$

$\therefore \text{fanout} = \frac{5.3636}{0.106} \approx 50.6 \approx 50 \text{ (floor)}$

$\frac{8.0 - 0V}{2} \times 5 = \frac{0V - (-12)}{5.5} \therefore (12 - 0) \times 5.36$

Case - II

$V_{OH} = 10V - 0.8V = 9.2V$

Supply current from driver $= \frac{12 - 10}{2.2} = 0.909 \text{ mA}$

Demand current $= \frac{10 - 0.8}{15} = 0.613 \text{ mA}$

$V_{BE} = 0.8V$
 $\therefore V_B = 0.8V$

$\text{fanout} = \frac{0.909}{0.613} = 1.48 \approx 1 \text{ (floor)}$

∴ Maximum possible fanout = $\min(50, 1)$
 $= 1$

(b)

$V_{OH} = 110V$

~~Driver is low, then load will be zero~~

$N=2$

(b) If driver input is low, then the input of load will be high.

Am 2828.2 = $\frac{5.0 - 5.1}{5.5} \times N = 2$ Supply current from battery

Am 201.0 = $\frac{12 - V_0}{2.2 \times 5.0} =$ Supply current

(root) 02 = $\frac{V_0 - 0.8}{15} =$ Individual demand

∴ $\frac{12 - V_0}{2.2} = 2 \times \frac{V_0 - 0.8}{15}$

⇒ $180 - 15V_0 = 4.4V_0 - 10.6$

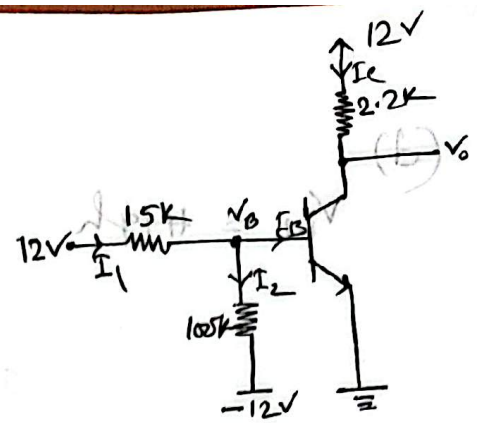
Am 000.0 = $\frac{0.1 - 5.1}{5.5} \Rightarrow 10.6V_0 = 181.6$

Am 12.0 = $\frac{V_0 - 0.8}{12} \Rightarrow V_0 = 9.36$

(root) 1 = $\frac{0.09 - 0}{8.12} =$ fanout

(c) If,

$V_{in} = \text{High}$



$$V_{BE} = 0.8V \text{ or } V_B = 0.8V$$

$$V_{CE} = 0.2V \text{ or } V_C = 0.2V = V_o$$

$$\therefore I_C = \frac{12 - 0.2}{2.2} = 5.36 \text{ mA}$$

$$I_1 = \frac{12 - 0.8}{15} = 0.746 \text{ mA}$$

$$I_2 = \frac{0.8 + 12}{100} = 0.128 \text{ mA}$$

$$I_B = I_1 - I_2 = 0.618 \text{ mA}$$

$$I_E = I_B + I_C = 5.97 \text{ mA}$$

$$A_{m 201.0} = \frac{12 + 12}{501 + 21} = 1$$

$$\therefore \text{Power dissipation} = (12 - (-12)) \times 0.128 + (12 - 0) \times 0.618 +$$

$$(12 - 0) \times 5.36$$

$$A_{m 200.0} = 1, A_{m 200.0} = 1, A_{m 200.0} = 1$$

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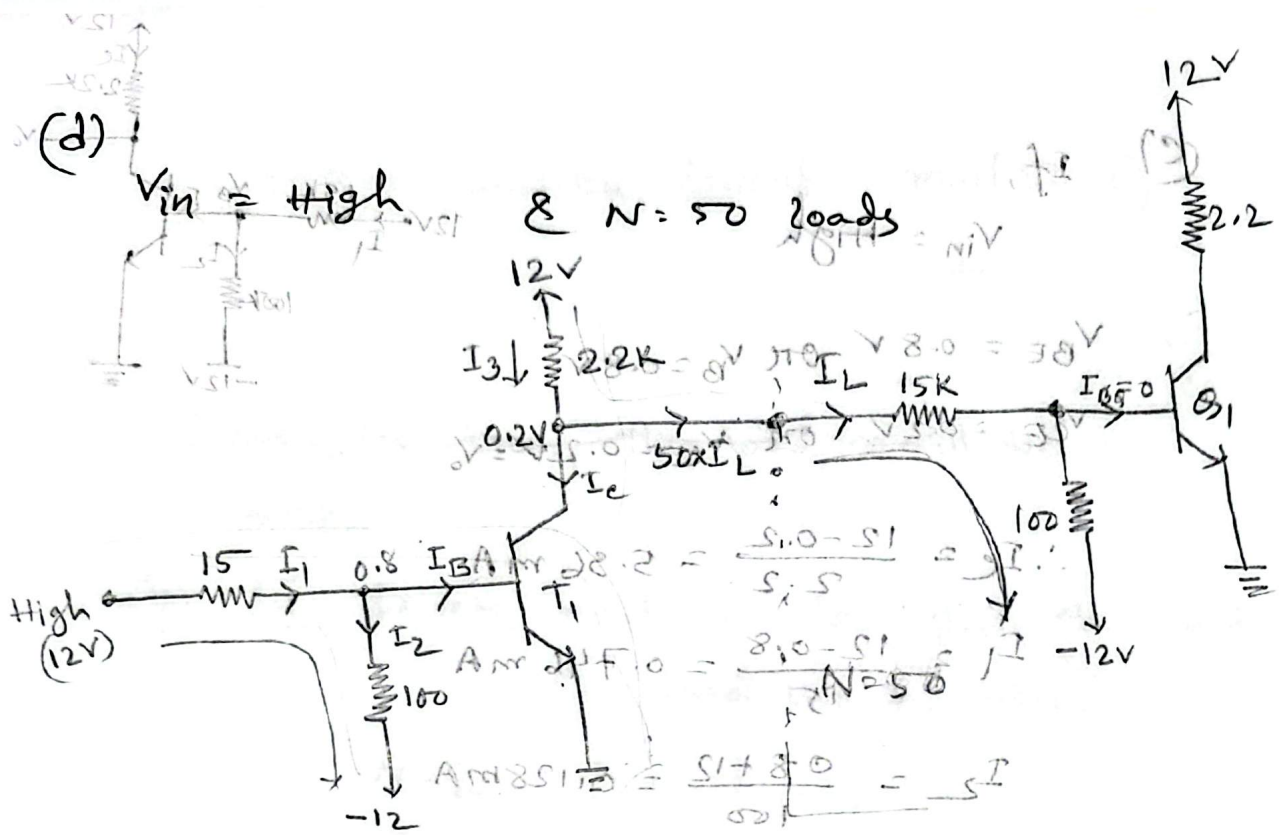
$$\text{Power dissipation} = (12 - (-12)) \times 0.128 + (12 - 0) \times 0.618 + (12 - 0) \times 5.36$$

$$= 12 \times (0.128 + 0.618 + 5.36)$$

$$= 12 \times 6.106$$

$$= 73.272 \text{ mW}$$

$$= 73.27 \text{ mW}$$



$T_1 \rightarrow \text{saturation}$ & $T_2 \rightarrow \text{cutoff}$

$$I_L = \frac{0.2 + 12}{15 + 100} = 0.106 \text{ mA}$$

$$I_3 = \frac{12 - 0.2}{2.2} = 5.363 \text{ mA}$$

$$I_1 = 0.746 \text{ mA} \text{ \& } I_2 = 0.128 \text{ mA}$$

$$I_B = 0.618 \text{ mA}, I_C = 0.063 \text{ mA}$$

Power Dissipation, $P = (12 + 12) \times I_2 + (12 - 0) \times I_B + (12 - 0) \times I_C$

$$+ 50 \times (12 + 12) \times I_L$$

$$= 138.444 \text{ mW}$$

(e) $V_{OH} = 10V$ & $V_{OL} = 0.2V$ $\therefore \frac{8.0 - 0.2}{12}$

$V_{10k\Omega} = 2.401V$

$V_{10k\Omega} = 2.401V$

V_{IL} calculation,

$V_{BE} = 0.5V$ or $V_B = 0.5V$

$I_B = I_C = I_E = 0$
 $V_{BE} = 0.5V$

$\frac{V_B - V_{IL}}{15} + \frac{V_B + 12}{100} = 0$

$\Rightarrow 100V_B - 100V_{IL} + 15V_B + 180 = 0$

$\Rightarrow 115V_B - 100V_{IL} = 100 \times 0.5 + 15 \times 0.5 + 180$

$\Rightarrow 100V_{IL} = 237.5$

$\Rightarrow V_{IL} = 2.375V$

V_{IH} calculation

$V_{BE} = 0.8V$ or $V_B = 0.8V$

$V_{CE} = 0.2V$ or $V_C = 0.2V$

$I_C = 5.36mA$ [from part (c)]

$\beta_F = 30$

$\frac{I_C}{I_B} = \beta_F$ or $I_B = \frac{I_C}{\beta_F} = 0.178$

$\therefore I_{Rin} = I_B + \frac{0.8 + 12}{100} = 0.306mA$

$$\therefore \frac{V_{in} - 0.8}{15} = 0.3065 \Rightarrow V_{in} = 5.401 \text{ V} \quad (2)$$

$$\Rightarrow V_{in} = 5.401 \text{ V}$$

$$\therefore V_{IH} = 5.401 \text{ V}$$

$$V_{NH} = 10 - 5.401 = 4.599 \text{ V}$$

$$N_{NL} = 2.375 - 0.2 = 2.175 \text{ V}$$

$$\text{Maximum Noise margin} = \min(4.599, 2.175)$$

$$0 = 0.81 + 0.4V_1 + 1.2V_2 + 1.80 = 0$$

$$0.81 + 2.0 \times 1.2 + 2.0 \times 0.2 = 1.00V = 1.00V$$

$$2.485 = 1.00V$$

$$V_1 = 5.342 \text{ V}$$

$$V_{IH} = 5.342 \text{ V}$$

$$V_{BE} = 0.8 \text{ V}$$

$$V_{CE} = 0.5 \text{ V}$$

$$I_C = 2.30 \text{ mA}$$

$$I_B = 0.2 \text{ mA}$$

$$R_{TH} = \frac{V_{TH}}{I_{TH}} = \frac{0.81}{0.2} = 4.05 \text{ k}\Omega$$

$$A_{mid} = \frac{V_{out}}{V_{in}} = \frac{V_{CE}}{V_{BE}} = \frac{0.5}{0.8} = 0.625$$

Ans to the ques no-2

(d)

(a)

If any input is logical low,

Let, $A = \text{Low}$

$$\therefore V_A = 0.1V$$

$\therefore D_1$ is ON

D_4 & D_5 turned OFF

$\therefore Q_1$ cutoff

$$\therefore I_{B1} = I_{C1} = 0$$

$$V_P = 0.1 + 0.7 = 0.8V$$

$$I_1 = \frac{10 - 0.8}{6k\Omega} \approx 1.533mA$$

Example 9.8.1

$$Amp\ 8.1 = \frac{5.5 - 0.1}{5} = 1I$$

$$I_D = I_1 = 1.533mA$$

$$Amp\ P.O = \frac{0 - 2.0}{5} = -1I$$

$$I_2 = 0$$

$$\therefore I_1 = 1.533mA, I_2 = 0, I_{B1} = 0, I_{C1} = 0, I_D = 1.533mA$$

$$Amp\ 2P.N = \frac{1.0 - 0.1}{5} = 10I$$

Power

Dissipation

$$P = (10 - 0.1) \times I_1$$

$$= (10 - 0.1) \times 1.533$$

$$= 15.177mW$$

(continued from p. 1)

$$Amp\ 0 = 0I, Amp\ 2P.N = 10I, Amp\ P.O = -1I, Amp\ 8.1 = 1I$$

$$20.4 \times 0.1 + 8.1 \times 0.1 = 10I \times (0 - 10) + 1I \times (0 - 0) = 9$$

$$Wm\ 2.52 = 10I$$

(b)

If any input is logical high,

Let, $A = B = C = \text{High}$.

$$\therefore V_A = V_B = V_C = 10V$$

D_1, D_2 & D_3 are turned off.

D_4 & D_5 are turned on

$\therefore Q_1$ is in Saturation mode

D_1 is off, $V_{BE} = 0.8V, V_{CE} = 0.2V$

$$\therefore I_D = 0mA$$

V_{D1}

$$V_B = 0.8V$$

$$V_P = 0.7 + 0.7 + 0.8 = 2.2V$$

$$I_1 = \frac{10 - 2.2}{6k\Omega} = 1.3mA$$

$$I_2 = \frac{0.8 - 0}{2} = 0.4mA$$

$$I_{B1} = I_1 - I_2 = 1.3 - 0.4 = 0.9mA$$

$$I_{C1} = \frac{10 - 0.1}{2} = 4.95mA$$

$$I \times \beta = \frac{I_{C1}}{I_{B1}} = \frac{4.95}{0.9} = 5.5 < \beta_F$$

$$I_1 = 1.3mA, I_2 = 0.4mA, I_{B1} = 0.9mA, I_{C1} = 4.95mA, I_D = 0mA$$

$$\text{Power Dissipation, } P = (10 - 0) \times I_1 + (10 - 0) \times I_{C1} = 10 \times 1.3 + 10 \times 4.95 = 62.5mW$$

(c)

Case - I

$V_y = \text{High}$

Then D_6 is turned OFF.

$$\therefore I_L = 0 \text{ mA}$$

$\therefore 0 \text{ mA}$ Current follow into driven circuit

~~$\therefore$ We can connect as many loads as possible~~

$$\therefore \text{Fanout} = \frac{\text{Supply}}{\text{Demand}} = \frac{\text{Supply}}{6} = \infty$$

Case - II

$V_y = \text{low} = 0.1 \text{ V}$

D_6 will be turning ON.

$$V_{\text{load}} = 0.1 + 0.7 = 0.8 \text{ V}$$

$$I_L = \frac{10 - 0.8}{6} = 1.533 \text{ mA}$$

$$I_3 = \frac{10 - 0.1}{2} = 4.95 \text{ mA}$$

$$I_{C\text{max}} = \beta_F \times I_{B1} = 30 \times 0.9 = 27 \text{ mA}$$

let, N numbers of loads are connected,

$$\therefore I_{C\text{max}} = I_3 + N \times I_L$$

$$\Rightarrow I_3 + N \times I_L = 27$$

$$\Rightarrow N = \frac{27 - 4.95}{1.533} = 14.383 \approx 14 \text{ (floor)}$$

$$\text{Fanout} = 14.$$

$$\text{maximum fanout} = \min(\infty, 14) = 14.$$

Ans.

Ans. to the ques no-3

Case I - 2200

$$V_{BE} = 0.7V$$

(a) $A = B = 5V$

770 current β of Δ mult

$$A_{MO} = \beta I$$

$\therefore D_A$ & D_B are turned off.

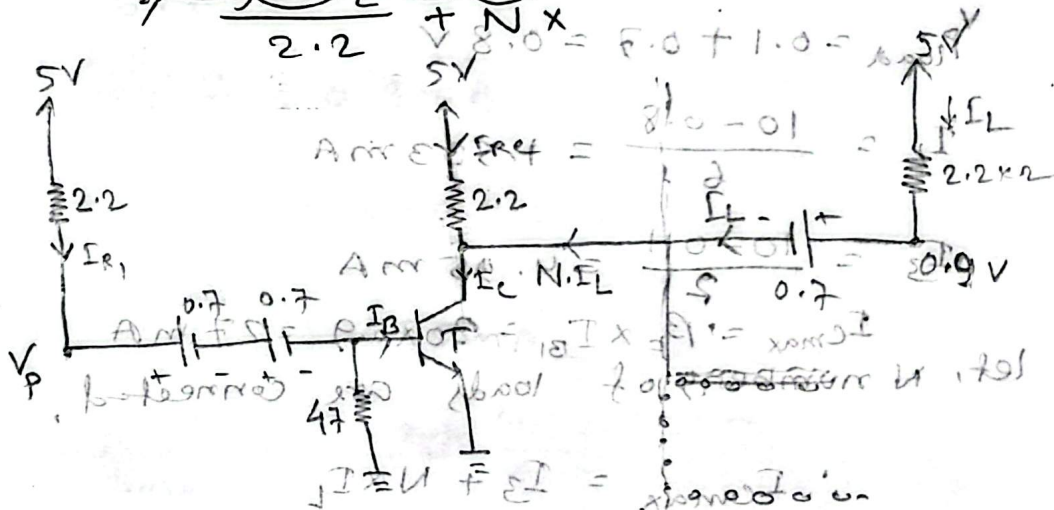
D_1 & D_2 are turned ON $\rightarrow T \rightarrow$ Saturation mod.

$$\therefore V_P = 0.7 + 0.7 + 0.8 = 2.2$$

$$\therefore I_B = \frac{5 - 2.2}{2.2} = \frac{0.8 - 0}{47} = 1.256 \text{ mA}$$

$$I_{C_{max}} = I_B \times \beta = 1.256 \times 65 = 81.62$$

$$I_{C_{max}} = I_{R_C} + N \times I_P$$



$$I_{R_C} = \frac{5 - 0.2}{2.2} = 2.18$$

$$I_L = \frac{5 - 0.9}{2.2} = 1.86$$

$$I = 1.86$$

$$I_{Cmax} = I_{EC} + N \times I_L$$

$$\Rightarrow 2.18 + N \times 1.86 = 81.62$$

$$\Rightarrow N = 42.70 \approx 42 \text{ (floor)}$$

$$\therefore \text{Maximum Fanout} = 42$$

(b)

$$A = 5V \text{ \& } B = 0V$$

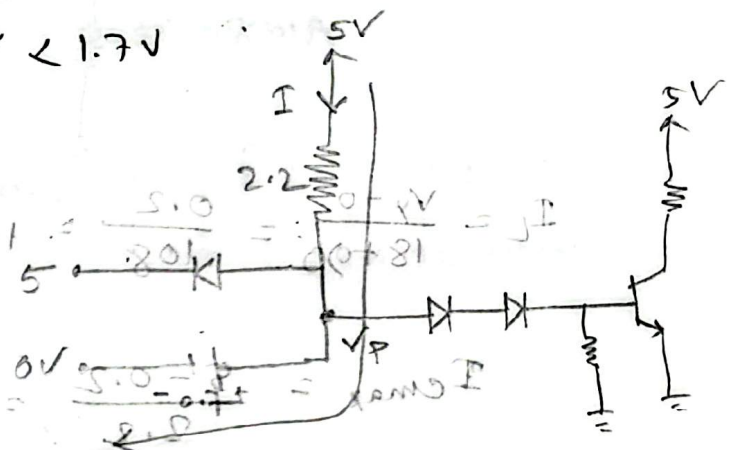
$$D_A \rightarrow R.B \text{ \& } D_B \rightarrow F.B$$

$$V_P = 0.7 + 0 = 0.7V < 1.7V$$

$\therefore T \rightarrow \text{Cutoff}$

$$\therefore I = \frac{5 - 0.7}{2.2}$$

$$A \approx 81.5 \text{ mA}$$



(note): Power Dissipation, $P = I(5 - 0) \times I$

$$(note) P_{F.H} = 28. P_{F.H} \times 1.954 = 9.77 \text{ mW}$$

$$J^T X N + J = \text{const}$$

$$52.18 = 23.1 \times 11 + 81.2$$

SP, 1st OF, SP 24 56

$$V_{in} = \frac{S.P.}{\text{transform multiplier}} = \frac{0.2}{1} = 0.2$$

5 ✓ ↑



5.5

Q. 120. 1.

1. 10/10/2020

$$120.1 \times 98.1179.35 \approx 1179 \text{ (floor)}$$

Was FF. e -

Ans to the ques no-4

$$I_C = \frac{V_{CC} - V_{CE}}{R_C} = \frac{5.0 - 0.5}{10} = 0.45 \text{ mA}$$

$$V_{in} = 0.2 \text{ V} = \frac{I_B R_B}{1} = \frac{I_B \cdot 10}{1} \Rightarrow I_B = 0.02 \text{ mA}$$

$$R = 10 \text{ k}\Omega$$

$$A_{mid} = \frac{V_{out}}{V_{in}} = \frac{5.0}{0.2} = 25$$

$$V_{in} = 0.2 \times 0.5$$

$$Q_1 \rightarrow \text{cutoff} \Rightarrow I_B + I_C = I_E$$

$$I_B = I_C = I_E = 0$$

$$I_1 = I_2 = I$$

$$\therefore I = \frac{0.2 - (-12)}{10 + 120} = 0.0938 \text{ mA}$$

$$\text{Power dissipation, } P = (0.2 + 12) \times 0.0938 = 1.14436 \text{ mW}$$

(b)

$$V_{NH} = 6 \text{ V}$$

$$V_{OH} = 10 \text{ V}$$

$$V_{OL} = 0.2 \text{ V}$$

$$V_{NH} = V_{OH} = V_{IH} = 10 \text{ V}$$

$$V_{IH} = 10 - 6 = 4 \text{ V}$$

$$Q_2 \rightarrow \text{Saturation}$$

$$V_{CE} = 0.2 \text{ mV, } V_{BE} = 0.8 \text{ mV}$$

$$V_{BE} = 0.8 \text{ mV, } V_B = 0.8 \text{ mV}$$

$$(V_{OH}) = 10 \text{ V}$$

$$\therefore V_0 = 0.2 \text{ V}$$

A - or loop wt of 2mA

$$I_C = \frac{12 - 0.2}{2.2} = 5.363 \text{ mA}$$

$$I_B = \frac{I_C}{\beta_F} = \frac{5.363}{30} = 0.1787 \text{ mA} \quad (1)$$

$$I_2 = \frac{0.8 + 12}{120} = 0.1067 \text{ mA}$$

$$\therefore I_1 = I_2 + I_B = 0.2854 \text{ mA}$$

$$\therefore \frac{V_{IH} - 0.8}{R} = 0.2854$$

$$R = \frac{11.212 \text{ k}\Omega}{0.51 + 0.1} = 11.212 \text{ k}\Omega$$

$$\therefore R = 11.212 \text{ k}\Omega$$

$$8200.0 \times (51 + 5.0) = 4.401 \text{ mA}$$

$$Wm d e p m i . l =$$

(c)

Case - I

$$V_{in} = \text{low} = 0.2 \text{ V}$$

$$V_2 = \text{HIV} \quad (d)$$

$$Q_1 = \text{cutoff}, Q_2 = \text{Saturation.}$$

$$V_0 = 10 \text{ V}, V_{B_2} = 0.8$$

$$\text{Max supply current} = \frac{12 - 10}{2.2} = 0.909 \text{ mA}$$

$$\text{Max demand} = \frac{10 - 0.8}{100} = 0.092 \text{ mA}$$

$$\text{Fanout} = \frac{\text{supply}}{\text{demand}} = \frac{0.909}{0.092} = 9.88$$

$$= 9 \text{ (floor)}$$

Case - II

$$V_{in} = 12V = \text{High}$$

$Q_1 \rightarrow \text{saturation}, Q_2 \rightarrow \text{cutoff}$

$$\therefore V_o = 0.2V$$

~~I_{Q2}~~

$$\text{Maximum Supply Current} = \frac{12 - 0.2}{2.2} = 5.3636 \text{ mA}$$

$$\text{Individual demand}_o = \frac{0.2 + 12}{100 + 120} = 0.0554 \text{ mA}$$

$$\text{Fanout} = \frac{\text{Supply}}{\text{Demand}} = \frac{5.3636}{0.0554} = 96.81$$

$$\approx 96 \text{ (floor)}$$

$$\text{maximum fanout} = \min(96, 9)$$

$$= 9$$